Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/01/2007
PI Name:	Barcellos-Hoff, Mary Helen Ph.D.		
Project Title:	NSCOR: Mechanisms of HZE Damage and Repair in Human Epithelial Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
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Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Former affiliation was Lawrence Berkeley National Laboratory; moved to NYU in August 2008. Moved to UCSF September, 2015		
Project Type:	Ground	**Solicitation / Funding Source:**	NSCOR 03-OBPR-02
Start Date:	12/01/2003	**End Date:**	05/29/2009
No. of Post Docs:	6	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes at that time, per JSC/A. Chu-ARC (jvp 4/2009) NOTE: NCE to 5/29/2009 has been approved, per A. Chu/ARC (10/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kronenberg, Amy (Lawrence Berkeley National Laboratory) Parvin, Bahram (Lawrence Berkeley National Laboratory) Chatterjee, Aloke (Lawrence Berkeley National Laboratory) Rydberg, Bjorn (Lawrence Berkeley National Laboratory) Yaswen, Paul (Lawrence Berkeley National Laboratory) Gray, Joe (Lawrence Berkeley National Laboratory) Costes, Syvlain (Lawrence Berkeley National Laboratory) Wyrobek, Andrew (Lawrence Berkeley National Laboratory)		
Grant/Contract No.:	NNA04CF75I		
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Performance Goal Text:	
Task Description:	NASA's Biomedical Critical Path Roadmap defines the carcinogenic risks of radiation exposure as one of only four Type I risks identified. Estimates of cancer risk from human travel in space should be based on a mechanistic understanding of complex effects that different types of radiation exposure elicit. Because most adult human solid tumors are epithelial in origin and because radiation effects are dictated by cellular genotype and phenotype, studies of HZE particle radiation biology using a physiologically relevant model will be the most informative regarding risk. We have developed methods for culturing human mammary epithelial cells (HMEC) derived from normal tissue as both monolayer and multicellular aggregates, which enable study of phenotypes and pathways that may be uniquely human, epithelial or multicellular. We have studied how controlled cellular and microenvironment conditions affect early responses to HZE particle radiation-induced DNA damage in terms of repair, foci formation and gene expression, and persistent radiation-induced phenotypes and pathways that precede neoplasia. The potential for inducing genomic instability, which may contribute to its carcinogenic effect, has been measured in both finite lifespan HMEC . Together, these surrogate endpoints will enable functional evaluation of neoplastic potential as a function of radiation quality, energy and dose. These data will be integrated at two levels: by theoretical modeling of the physical events leading to DNA damage and by systems biology modeling of critical pathways. The intent of this 5-year project NSCOR at Lawrence Berkeley National Laboratory Life Sciences Division is to provide a comprehensive examination of HZE effects from the initial damage, to early cellular HMEC responses, to persistent functional precursors of carcinogenesis. We have addressed the mechanisms, nature, and frequency of DNA damaging events; mechanisms of DNA repair and misrepair, early signal transduction mechanisms; immediate and long-term, reversible and irreversible gene expression changes; cellular remodeling and reorganization; potential mechanisms of tissue "repair" and matrix effects; and the cellular and molecular mechanisms of HZE/proton induced progression to a neoplastic phenotype.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Current modeling of risk relies on comparing the relatively biological effectiveness of low LET radiation to high LET radiation effects, thus providing an earth benefit in the description and understanding of low dose reference radiation effects. In the context of the NSCOR, we have developed image analysis programs to quantify DNA damage response (e.g. radiation-induced foci) and specific organelles (i.e. centrosomes) using high-throughput imaging bioinformatics. The characterization of non-malignant human mammary epithelial cells responses to ionizing radiation provides mechanistic insights to the processes relevant to human health risks from diagnostic and therapeutic radiation exposures. The NSCOR has supported the development of a new media that enables human mammary epithelial cells to be maintained longer in culture.
	We have conducted two experimental runs at BNL-NSRL using the 1 Gev/amu Fe beam in the last year to test how radiation responses of human mammary epithelial cells are affected by radiation quality. Epithelial cells depend on signals from the microenvironment to establish the requisite polarity for functional differentiation. Because most adult human solid tumors are epithelial in origin and because radiation effects are dictated by cellular genotype and phenotype, studies of HZE particle radiation biology using a physiologically relevant model will be the most informative regarding risk. We developed methods for culturing human mammary epithelial cells (HMEC) derived from normal tissue as both monolayer and multicellular aggregates, which enables the study of phenotypes and pathways that may be uniquely human, epithelial or multicellular. We have examined how controlled cellular and microenvironment conditions affect early responses to HZE particle radiation-induced DNA damage in terms of repair, foci formation and gene expression, genomic instability, and persistent radiation-induced phenotypes and pathways that precede neoplasia. The LBNL NSCOR is studying these surrogate endpoints with the goal evaluating neoplastic potential as a function of radiation quality, energy and dose. We hypothesized that HZE particles such as 1GeV/amu Fe-ions induce more persistent DNA damage than sparsely ionizing gamma-rays. To address this, we measured gamma-H2AX and 53BP1 radiation-induced foci (RIF) in proliferating HMEC after exposure to low doses of sparsely and densely ionizing radiation. Preliminary results revealed the disappearance of radiation induced gamma-H2AX and 53BP1 RIF in HMEC have different dynamics as a function of radiation quality. While RIF numbers after gamma-ray irradiation returned to control level after about 12h, Fe-ion induced foci were still present at 24h. However, these persistent RIF were removed during the 24 – 48h time interval after irradiation under the conditions used. The slower disappearance of foci after Fe-ion exposure could be due to the fact that HZE particles induce more complex DNA damage in the core of the ion track, which will be tested in future experiments using different ions and LETs. Since energy is randomly deposited along high-LET particle paths, RIF along these paths should also be randomly distributed. The probability to induce DSB can be derived from DNA fragment data measured experimentally by pulsed-field gel electrophoresis. We used this probability in Monte Carlo simulations to predict DSB locations in synthetic nuclei geometrically described by a complete set of human chromosomes, taking into account microscope optics from real experiments. As expected, simulations produced DNA-weighted random (Poisson) distributions. In contrast, the distributions of RIF obtained as early as 5 min after exposure to high LET (1 GeV/amu Fe) were non-random. This deviation from the expected DNA-weighted random pattern can be further characterized by “relative DNA image measurements”. This novel imaging approach shows that RIF were located preferentially at the interface between high and low DNA density regions, and were more frequent than predicted in regions with lower DNA density. The same preferential nuclear location was also measured for RIF induced by 1 Gy of low-LET radiation. This deviation from random behavior was evident only 5 min after irradiation for phosphorylated ATM RIF, while gammaH2AX and 53BP1 RIF showed more pronounced deviations up to 30 min after exposure. These data are now in press in PLoS Computational Biology. While to date, many studies of ionizing radiation have focused on its effects on primary DNA structure, comparatively few have focused on its effects on epigenetic phenomena such as gene methylation or phenotype. Such epigenetic phenomena have been increasingly implicated in carcinogenesis in human cells and tissues, and in some cases, may

Task Progress:	represent predominant mechanisms of oncogenesis. In serum-free growth medium, HMEC from histologically normal breast tissue growth arrest after 5-20 population doublings, exhibiting senescent morphologies and elevated expression of p16. Epigenetic silencing of the cyclin-dependent kinase inhibitor p16INK4a (p16) in human cancers is often associated with methylation of the p16 promoter by unknown mechanism(s). Such HMEC cultures spontaneously yield at low frequencies (1x10-5 to 1x10-8) variant p16(-) cell populations that are capable of long term growth (50-100 total population doublings), and which are susceptible to genomic instability associated with telomere dysfunction. We sought to determine whether radiation would alter p16 expression, p16 promoter methylation, and/or long term growth potential in HMEC. In replicate experiments with HMEC from three individuals, we determined that 2 Gy X-irradiation causes increases in the appearance of p16(-) cells with long term growth potential. Flow cytometry confirmed that the differences in BrdU incorporation in the treated and untreated populations were statistically significant (P = 0.0016). Based on these data, we hypothesize that radiation to cause epigenetic changes in p16 expression. We are currently exploring the mechanistic underpinnings of this phenomenon. We have demonstrated that irradiated pre-malignant HMEC, cultured in reconstituted basement membrane and transforming growth factor beta1 (TGFbeta), fail to undergo tissue-specific morphogenesis when embedded in reconstituted basement membrane, which is strongly associated with malignancy and deregulation of differentiation. Our recent studies show that disrupted morphogenesis is a result of irradiated cells undergoing TGFbeta mediated epithelial to mesenchymal transition (EMT). These data are now in press in Cancer Research. We compared eqi-toxic doses of 1Gy of 1 GeV/amu 56Fe to 2Gy of 137Cs gamma-radiation in both monolayer and 3D culture. Both exposures resulted in EMT in monolayer and disrupted morphogenesis when cells were cultivated in the presence of TGFbeta. Radiation dose responses show that this effect has a very low threshold in that a single exposure of 3-100 cGy gamm-radiation elicits the ‘same’ phenotypic switch. Fe ions were more toxic than x-rays to HMEC irradiated in growth arrested monolayer cultures, with a RBE of 1.8 at the D(10) level with a delayed plating protocol. Irradiation as growth arrested acinar structures protected HMEC against x-ray-induced cell death. In contrast, the 3D architecture did not protect the HMEC from killing by Fe ions. Exposure of HMEC in growth arrested monolayer culture to x-rays or Fe ions elicited severe karyotypic instability, defined as at least 3 subpopulations of cells per colony, in the clonal descendants of ~20% of clones. This phenotype was not seen in any of the unirradiated colonies. Severe karyotypic instability was seen in colonies arising after exposure to the lowest dose of Fe ions (0.5 Gy), but no evidence of severe karyotypic instability was seen for colonies arising after exposure to <2 Gy of x-rays. There was no evidence of a dose response for severe karyotypic instability. These data are now submitted. A dose response was seen for centrosome aberrations (CA) defined as 3 or more centrosomes/cell for the progeny of cells exposed in monolayers to x-rays or Fe ions. Unirradiated clones had a baseline frequency of 0.92% CA. The dose response appeared to saturate with ~4% of cells/colony exhibiting this phenotype. There was no evidence for an RBE>1 for the induction of centrosome hyperamplification following irradiation in 2D. However, irradiation in rBM reduced the level of centrosome abnormalities after x-irradiation, but did not protect against Fe ion exposure, suggesting that there may be an increased RBE for centrosome abnormalities in cells exposed to Fe ions. Studies have analyzed gene expression induced by low-LET radiation in growth arrested HMEC using statistical and bioinformatics approaches with the goal of identifying specific signaling pathways. Image analysis continues to develop strategies that can be used to characterize the diversity of cell responses and track the development of phenotypes over time. Together these studies show that in general high vs low LET radiation is more effective in eliciting many of these endpoints. We propose that these studies warrant further examination of whether individual phenomena (e.g. damage response, phenotype, genomic instability) interact in a unique manner to drive neoplastic transformation following high LET exposure. We have begun a new project in collaboration with the Boston NSCOR to model these events at the cellular level.

Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Parvin B, Yang Q, Han J, Chang H, Rydberg B, Barcellos-Hoff MH. "Iterative voting for inference of structural saliency and characterization of subcellular events. " IEEE Trans Image Process. 2007 Mar;16(3):615-23. PMID: 17357723 , Mar-2007
Articles in Peer-reviewed Journals	Andarawewa PK, Erickson AC, Costes SV, Bissell MJ, Barcellos-Hoff MH. "Ionizing radiation predisposes human mammary epithelial cells to TGF-ß induced epithelial to mesenchymal transition." Cancer Research, In press, August 2007. Expected publication November 2007. , Aug-2007
Articles in Peer-reviewed Journals	Costes SV, Ponomarev A, Chen JL, Nguyen D, Cucinotta FA, Barcellos-Hoff MH. "Image-based modeling reveals dynamic redistribution of DNA damage into nuclear sub-domains." PLoS Compuational Biology, 3(8): e155. Published online August 3, 2007. http://dx.doi.org/10.1371/journal.pcbi.0030155 , Aug-2007
Articles in Peer-reviewed Journals	Barcellos-Hoff MH. "Systems radiation biology: Cancer as an emergent phenomenon." International Journal of Radiation Biology, In press, August 2007. Expected publication November 2007. , Nov-2007
Articles in Peer-reviewed Journals	Kirshner J, Jobling MF, Pajares MJ, Ravani SA, Glick AB, Lavin MJ, Koslov S, Shiloh Y, Barcellos-Hoff MH. "Inhibition of TGFß1 signaling attenuates ataxia telangiectasia mutated activity in response to genotoxic stress." Cancer Res. 2006 Nov 15;66(22):10861-9. Epub 2006 Nov 6. PMID: 17090522 , Nov-2006
Articles in Peer-reviewed Journals	Raman S, Maxwell CA, Barcellos-Hoff MH, Parvin B. "Geometric approach to segmentation and protein localization in cell culture assays." J Microsc. 2007 Jan;225(Pt 1):22-30. PMID: 17286692 , Jan-2007
Books/Book Chapters	Andarawewa KL, Kirshner J, Mott JD, Barcellos-Hoff MH. "TGF beta: Roles in DNA Damage Responses." in "Transforming Growth Factor-Beta in Cancer Therapy, Volume 1 Basic and Clinical Biology, 1st edition." Ed. S.B. Jakowlew. Totowa, NJ : Humana Press, 2007., Jun-2007

Papers from Meeting Proceedings	Chang H, Andarawewa PK, Han J, Barcellos-Hoff MH, Parvin B. "Perceptual grouping of membrane signals in cell-based assays." IEEE International Symposium on Biomedical Imaging: from Nano to Macro, Arlington, VA, April 2007. IEEE International Symposium on Biomedical Imaging: from Nano to Macro, 2007, pp. 532-536. ISBN: 1-4244-0672-2 http://dx.doi.org/10.1109/ISBI.2007.356906 , Apr-2007
Papers from Meeting Proceedings	Han J, Chang H, Yang Q, Barcellos-Hoff MH, Parvin B. "Segmentation of mammosphere structures from volumetric data." IEEE International Symposium on Biomedical Imaging: from Nano to Macro, Arlington, VA, April 2007. IEEE International Symposium on Biomedical Imaging: from Nano to Macro, 2007. pp. 524-527. ISBN: 1-4244-0672-2 http://dx.doi.org/10.1109/ISBI.2007.356904 , Apr-2007
Papers from Meeting Proceedings	Chang H, Parvin B. "Segmentation of three-dimensional cell culture models from a single focal plane. " Second International Symposium, ISVC 2006 Lake Tahoe, NV, USA, November 6-8, 2006. Berlin : Springer, 2006. pp. 586-595. (Advances in Visual Computing ; vol. 4292) http://dx.doi.org/10.1007/11919629_59 , Nov-2006